Traditional Agriculture in Southeast Asia

A Human Ecology Perspective

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Preface - Gerald G. Marten

The concept for this book was conceived by the East-West Center Working Group on the Human Ecology of Traditional Agroecosystems, which convened at the East-West Environment and Policy Institute in June-August 1982. The group consisted of agronomists, ecologists, and social scientists from the United States and the Southeast Asian Universities Agroecosystem Network (SUAN). The working group set out to develop a conceptual framework for describing how traditional agriculture functions and to translate that framework into operational procedures for collaborative research on traditional agriculture that would involve both natural and social scientists. The ecosystem concept, in this case the agricultural ecosystem, served as a starting point.

The scientists from Southeast Asia were interested in ecological research because of problems that had arisen with new cropping systems being developed in their countries. The problems included soil degradation and crop losses due to drought, pests, and diseases. Some of these problems were particularly alarming because they became apparent only after a new cropping system was employed for a number of years. The scientists felt a growing concern about the risks inherent in promoting new cropping systems that had not yet proved themselves under local conditions.

There was also a need to incorporate social science into cropping systems development. Farmers often were not using new cropping systems, even though the new systems appeared to provide higher yields and higher economic returns than the existing agriculture. Because the new cropping systems were apparently not what farmers needed, the scientists concluded that efforts to develop better cropping systems would be enhanced by a better understanding of how agricultural households function and make their farming decisions.

As the scientists proceeded to address the ecological and social dimensions of local agriculture, they came to appreciate how well adapted it was to local conditions and how much of this could be attributed to traditional practices that had evolved over many generations. As a consequence, they were interested in drawing upon both traditional agricultural technology and modern agricultural science for new cropping system development. This interest was stimulated further by concern for the rapid rate at which traditional agricultural practices and knowledge appeared to be disappearing under the impact of modernization.

Some of the U.S. scientists wanted to learn about traditional agriculture because of their interest in exploring alternatives for modem U.S. agriculture. Although U.S. agriculture is highly productive per unit of labor, it is also dependent on heavy inputs of energy and petrochemicals that could become scarce. U.S. agriculture also has problems of environmental pollution and soil degradation that raise doubts about its long-term viability in its present form. Because the self sufficiency and centuries of sustainable use of much of traditional Southeast Asian agriculture are strong in ways that U.S. agriculture is not so strong, the traditional agriculture may provide a source of ideas for alternatives. While it is unlikely that many of the details of traditional agriculture from a tropical setting would apply to the very different environmental and social conditions in the United States, some of the principles on which traditional agriculture is organized could be of more universal application.

The working group reviewed existing knowledge on the ecology of traditional agriculture, drawing upon the scientific literature and personal experience. The inquiry was organized around the following series of questions:

Agroecosystem Structure and Function

- What are the major types of traditional cropping systems in Southeast Asia?
- How do the types vary in their spatial and temporal structure?
- What are their design characteristics with respect to species richness and vertical and horizontal structure?
- How does the structure of traditional agroecosystems relate to their productivity, stability, and sustainability?

Soils

- How does the root stratification of mixed cropping systems contribute to full utilization of soil moisture and soil nutrients?
- What kinds of soil problems do small-scale farmers face in the tropics? How do farmers recognize the problems? What are their strategies for overcoming them?
- How do soil constraints determine the cropping systems that are employed? Do mixed cropping systems offer advantages on variable or marginal soil types? Do they reduce nutrient depletion on poor soils?
- What are the impacts of mixed cropping agriculture on erosion?
- **Pests**
- What are the ecological mechanisms of pest control in traditional agriculture? How does the temporal and spatial diversity of crops in traditional mixed cropping systems limit pest damage? How are pest problems affected by the mosaic of agroecosystems in a rural landscape?

- How do farmers define acceptable limits of pest damage to their crops? What are beneficial roles of "pests"?

Decision Framework of Traditional Farmers

- How do agricultural and social systems interact in a traditional farming society?
- What is the role of traditional knowledge in agroecosystem management strategies? How do traditional farmers perceive ecological factors that bear upon their management decisions? How do farmers adapt their cropping systems to local conditions (ecological and social)?
- How is traditional agriculture changing as a consequence of contemporary changes in society? What factors influence the acceptance or rejection of new technology by traditional farmers?

Although the group was not able to answer many of the questions fully, it did make tangible progress toward its objectives. Group members concluded that understanding how traditional agriculture functions can be greatly facilitated by a human ecology perspective, a perspective that focuses on the interactions between small-scale farmers and the farm ecosystems (i.e., agroecosystems) on which they depend for a living. This perspective embraces the total system, including not only the biophysical agroecosystem but also the human social system that shapes the organization of an agroecosystems soil, water, crops, and pests to generate desired outputs of goods and services. The human ecology perspective gives attention not only to the immediate consequences of changing agricultural practices but also to long-term consequences as the effects of a change reverberate through the biophysical ecosystem and between the ecosystem and the human social system.

This book presents the working group's results. Some of the chapters are composites of what different members of the group, with their different backgrounds, had to offer. Some of the chapters were augmented, and other chapters added, with contributions from other scientists in the Southeast Asian Universities Agroecosystem Network and the East-West Environment and Policy Institute.

An ultimate goal of research on traditional agroecosystems is to help agricultural scientists utilize traditional agricultural technology in strengthening the development process. This book is a step in that direction. It describes how traditional agriculture functions, conveys its strengths and weaknesses, and suggests the kind of role it can have in a changing world. There is no attempt to represent human ecology research on traditional agriculture as a replacement for existing modes of agricultural research nor to layout recipes for research. The members of the working group developed methods for collaborative agroecosystem research among themselves, but they did not attempt to develop systematic methodologies for designing better agricultural systems, although this was an ultimate concern. There is already an effective array of agricultural research methods under the rubric of "farming systems," which aim at developing new agricultural technologies for small-sale farmers. As with human ecology, the perspective of farming systems research is holistic, and many of the motivations for farming systems research are similar to those that led to this book. Although this book does not attempt to duplicate what a number of books on farming systems have already presented so well, many of the ideas should serve to augment the farming systems approach.

Acknowledgments - Gerald G. Marten

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Introduction - Gerald G. Marten

This book describes how traditional agriculture in Southeast Asia functions. It attempts to convey an appreciation for the sophistication of traditional agriculture, explaining why traditional agriculture has been so effective at meeting the needs of small-scale farmers for so many years. The role of traditional agriculture in today's changing world is explored in terms of ecological and social realities of contemporary agricultural development. Many of the details in the book are drawn from Indonesia, Thailand, and the Philippines, but the general picture applies to much of the rest of Southeast Asia and to many other areas of the tropics as well.

A number of themes recur throughout the book. One is the importance that small...scale farmers attach to considerations beyond high crop yields when deciding how to manage their agriculture. Considerations such as stability (the reliability of the yield), sustainability (the prospect for maintaining it in the long term), resilience (the ability to function under unexpected and possibly disastrous changes in social or environmental conditions), and efficiency (with respect to capital and labor inputs) are essential for survival.

A related theme is the complexity and elegance of traditional farm ecosystems (i.e., agroecosystems) and their intricate coadjustment with the social systems of the farmers who create them. One of the striking features of many traditional agroecosystems is their resemblance to natural ecosystems, a trait that contributes substantially to their stability, sustainability, resilience, and efficiency by following "nature's strategy" instead of struggling against it. Much of the unique character of traditional agroecosystems lies in the fact that like natural ecosystems they possess a highly structured diversity based on numerous plant species. This diversity fills a variety of subsistence needs, giving small...scale farm households substantial self...sufficiency and security while providing a flexibility that allows farmers to adapt their agriculture to uncertain and fluctuating social and environmental conditions.

Finally, there is a theme of change. In addition to changes associated with a fluctuating environment, there are changes due to a worldwide effort to modernize tropical countries through Western technology and new opportunities for consumption and involvement in a cash economy. These changes, combined with fundamental changes in the relations of subsistence farmers to their land as a consequence of the population explosion, have forced small-scale farmers in Southeast Asia to adjust their agriculture. This is not to suggest that traditional agriculture was static in the past. Southeast Asia has experienced numerous sweeps of foreign cultures and technologies over the millennia, and what we call traditional agriculture stems from a continuous process of assimilation and adjustment. Today we are witnessing that process in rapid progression-traditional and modem agriculture functioning side by side-each with its particular strengths and weaknesses and often blended within a single household's farming activities. By understanding the role of traditional agriculture in this process, it may be possible to clarify the role it should have in the future.

The first two chapters set the scene. Chapter 1 provides an overview of the major agricultural systems in Southeast Asia-for example, rice paddies, shifting cultivation, homegardens, commercial temperate vegetables, and commercial tree plantations-relating their organization in space and time to the environmental and social conditions with which they are associated. Each of the agricultural systems described in Chapter 1 is common in its occurrence throughout Southeast Asia, but each takes on a diversity of forms in different parts of the region. Most rural landscapes in Southeast Asia are a patchwork of a number of these agricultural systems, and even a single agricultural household often earns its living by simultaneously employing several of these agricultural systems on the different parcels of land at its disposal or on the same land at different seasons of the year. The character of these agricultural systems as practiced in different parts of Southeast Asia can vary enormously from purely traditional to heavily influenced by Western technology. Some of the systems, such as homegardens and other forms of mixed cropping, tend to be more traditional and therefore receive considerable attention throughout the book. The traditional component of other systems, such as commercial vegetable farming and commercial tree plantations (e.g., rubber or oil palm), tends to be less significant, and those systems receive correspondingly less attention.

Chapter 2 sets the conceptual scene by outlining the human ecology perspective that embraces not only the farm as a biophysical ecosystem but also the human social system that interacts with that ecosystem in many ways. On the one hand, a human ecology perspective is concerned with the structure and function of agricultural ecosystems-how the soil, water, crops, pests, and other biological and physical components of the ecosystem interact to generate both desired and undesired outputs of goods and services. On the other hand, a human ecology perspective is concerned with how small-scale farmers interact with the agroecosystems on which they depend for a living and how the human social system shapes the interactions between farmers and their agroecosystems.

The framework in Chapter 2 is expressed in several different ways through the rest of the book. Chapters 3 to 6 present detailed examples of traditional agriculture in the Philippines, Thailand, and Indonesia to give a comprehensive picture of traditional agriculture, the organization of crops in space and time, the management of other parts of the agroecosystem (e.g., soil and pests), and interactions of the human social system with the biophysical agroecosystem. Chapters 7 to 9 focus on various aspects of the social system and how they interact with traditional agroecosystems, while Chapters 10 to 13 each focus on a single agroecosystem element (e.g., crop pests) from the perspective of that element's interactions with the rest of the agroecosystem and with the social system. Chapter 14 takes a "slice" through these two complex systems by following a single topic (nutrition) through the numerous points of interaction between agroecosystem and social system.

Chapter 3 describes the highly traditional wet rice and swidden agriculture in a remote, mountainous area of the Philippines. It shows how custom and ritual are used to allocate land and water resources among agricultural households, to organize and coordinate village labor around an agricultural calendar that interdigitates several different cropping systems through the year, to cope with environmental hazards (e.g., typhoons and pests) that threaten the crops, and to sustain the productivity of their agriculture generation after generation. Chapter 3 also describes how integration with the broader Philippine community is changing the way these people interact with their agricultural ecosystems.

Chapter 4 contrasts the agriculture in two parts of Northern Thailand. The first is more traditional, in a mountainous area where the main cropping systems are upland rice" and swidden. The second area is a large irrigated valley where conditions are generally favorable for agriculture; rice and other crops can be grown through most of the year. The valley area is fully integrated with a market economy, has been thoroughly exposed to modem agricultural technology, and emphasizes temporal sequences of monocultures. Nonetheless, the area retains much of its traditional agriculture. Chapter 4 describes the interplay of traditional and modem agriculture.

Chapter 5 describes the rainfed agriculture in Northeast Thailand where soils are poor and rainfall is marginal and erratic. Traditional agriculture appears to have made the most of available resources. Different cropping systems are adjusted to subtle differences in land, and farm households coordinate their labor among the different cropping systems. This chapter relates recent efforts of scientists to intensify the agriculture in Northeast Thailand and the difficulties they have encountered in developing cropping systems with performance better than the traditional ones.

Chapter 6 describes two highly sophisticated mixed cropping systems that account for most of the food production in addition to rice in Java, where a high human population density forces farmers to extract their subsistence efficiently from a small amount of land. Perennial tree crops are an important part of both cropping systems. In the homegarden, trees are interplanted with field crops; in the other system, tree plantations are rotated with field crops to form a highly intensive "permanent swidden."

Chapter 7 introduces the social science perspective by outlining factors that influence how small-scale farmers make their farming decisions. This chapter points out that the interaction of farm households with the larger social system includes not only relations between households in a village (e.g., land tenure) but also interactions of the village with the outer world. Cultural preferences, new agricultural technology, off farm employment opportunities, rural...urban migration, expanding agricultural markets, agricultural commodity prices, expanding opportunities for consumption, and the actions of governments all influence the crops a household decides to produce and how it produces them.

Chapter 8 continues the social science perspective by examining some of the relations between traditional agroecosystems and human social organization. It starts with a historical survey of ideas on how the social organization of agrarian societies is associated with the environmental conditions under which they pursue their agriculture. Chapter 8 then reviews examples that illustrate interrelations between agroecosystems and several important elements of social organization such as land tenure, population structure, ethnic identity, ritual beliefs, and the organization of production.

The technology that forms the basis for structuring traditional agroecosystems is codified in the form of knowledge passed from generation to generation. Chapter 9 explores the nature of this traditional knowledge within the rubric of the science of "ethnoecology." The chapter is concerned with how traditional farmers perceive the agroecosystems they are managing and how this translates into their management practices.

Chapter 10 moves to the biophysical side of agroecosystem function by focusing on soil processes. It attempts to show how traditional agriculture has managed to persist on the same land for so long without the kind of soil degradation that creates a dependence on fertilizer applications to maintain yields. Traditional agriculture has employed a variety of strategies to maintain soil fertility, including the use of mulch and other crop residues to maintain soil organic matter, the recycling of mineral nutrients to maintain nutrient levels in the agroecosystem, and the maintenance of vegetative cover on the soil to prevent soil erosion. The mixed cropping character of traditional agriculture, which has had a major role in all of these functions, is examined in particular ecological depth. Also of importance is the detailed knowledge that traditional farmers have about the capabilities of different kinds of land and the ability of farmers to match different cropping systems with different kinds of land.

Chapter 11 continues with soils in the specific context of shifting cultivation. By looking at the changes in soil quality that occur when a tropical soil is cultivated for a number of years in succession, as well as the role of the fallow in replenishing and sustaining soil quality, Chapter 11 offers insights into the sustainability of tropical agriculture as it becomes more intensified due to increasing human population pressure.

Chapter 12 addresses the ways that traditional farmers deal with weed and insect pests in their agroecosystems. Traditional agriculture is noteworthy in this respect because serious pest damage is unusual even though many kinds of pests are found in traditional fields and modern chemical pesticides are not employed. What are traditional farmers doing right? The answer is complex, but it appears to lie in a combination of the pest resistance of local crop varieties, the fact that a number of different crops are often interplanted in the same field, and the fact that traditional agricultural landscapes are usually a patchwork of different cropping systems.

Chapter 13 describes the application of a rapid field assessment methodology to a single agroecosystem topic, the role of trees in rice fields of Northeast Thailand. This chapter takes a temporal perspective, describing how the numbers and kinds of trees change during the years following the clearing of a forest for rice paddy. The trees serve a multiplicity of functions building materials, fuelwood, mulch, food-but also must be kept in their proper place to avoid excessive interference with the rice crop.

Chapter 14 illustrates how detailed numerical data from a field study can be used to explore the interplay of ecological and social factors around a single agroecosystem topic, the integration of three agricultural systems in Java to provide a balanced diet. The chapter describes the impact of household economic status, land tenure, dietary preferences, and involvement' in a cash economy on the cropping systems a household employs and the consequences for nutrition. Of particular significance is crop diversity, which is highly developed in Javanese agriculture but which may be eroded as the agriculture modernizes.

Chapter 15 discusses the implications of recent agricultural development policies in Southeast Asia for agricultural research in general and human ecology research on traditional agriculture in particular. Any serious attempt to improve the agriculture of Southeast Asia on a broad geographical scale will require an attention to local environmental and social conditions beyond the centralized agricultural research and development strategy that has prevailed until recently. It may be necessary for agricultural research to draw on the intellectual resources of small-scale farmers in a variety of ways, not only identifying traditional agricultural practices of particular value but also incorporating farmers as equal partners with scientists in the agricultural research process.